Iayan Ansari

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Education

Rose-Hulman Institute of Technology, Terre Haute, IN May 2027

B.S. Mechanical Engineering & Computational Science (Double Major) | GPA: 3.08/4.00

Concentration in Computational Mechanics; Minor in German Language

Relevant Coursework: FEA, CFD, Machine Component Design, Design for Manufacturing, Numerical Methods

Honors: Finalist (top 10 of 100+ teams), Stanford Health Design Competition, Oct. 2025

Internship Experience

Chemical Engineering Intern – Hyera Inc. June 2026 – Aug. 2026

- Developing a coupled CFD-DEM model (ANSYS Rocky–Fluent) of steam activation in a double-walled rotary kiln by extracting the flighted fluid domain from plant CAD; meshing in progress.
- Built a kiln temperature data acquisition system logging 4 thermocouple channels with a HOBO UX120-014M logger, providing validation data for the CFD model.

Chemical Engineering Intern – Hyera Inc. July 2025 – Aug. 2025

- Maximized predicted activated-carbon yield over 5 process variables in 300 evaluations by building a 1D MATLAB reactor model, calibrating it to kiln trials, and applying GA/surrogate optimization.

Projects

6-Axis Robotic Arm for ASTM D4607 Testing | Software & Controls Sept. 2026 – Present

- Leading controls software for a capstone 6-axis arm automating iodine-number testing for client Hyera Inc., targeting 8+ h of unattended runtime, while co-designing its mechanism.
- Trained and simulated the arm's control policies in NVIDIA Isaac Sim before physical deployment, validating motions in a digital twin of the lab workcell to de-risk hardware integration.
- Built a computer vision spill detector recognizing 5 lab-vessel classes at 0.771 mAP50 and 10.9 ms/frame by training YOLO11s on 2,425 images, so the arm halts on spills.

Bike Crank-Arm Interface FEA | ME 422 (Team of 2) Winter 2026

- Showed a splined Al 7075-T6 crank-arm interface carries ~20% lower peak stress (396 vs. 498 MPa) than a tapered one by running 2-step ANSYS stress analysis with bolt preload and a 4.15 kN load.
- Verified the method within 1% of analytic stresses on benchmark shafts and ran mesh convergence studies to <2.3% change by refining elements to 0.56 mm at the bolt holes.

Physics-Informed Neural Networks vs. Classical Solvers | ME 427 Spring 2026

- Showed PINNs plateau at 0.9–2.7% L2 error on the shock-forming Burgers equation, over 1,000x worse than a 4th-order ETDRK4 reference, by benchmarking 4 PyTorch PINN variants.
- Achieved the lowest max error of 4 PINN variants (0.048 vs. 0.065 vanilla) by adding residual-based adaptive refinement of collocation points near the shock.

Leadership Experience

President – Muslim Student Association Apr. 2024 – Present

- Delivered khutbahs (religious lectures) about topics in Islam and modern humanitarian crises.
- Assist the local mosque with cleaning, visitations, and preparing food during the month of Ramadan.

Treasurer – South Asian Student Alliance Apr. 2025 – Present

- Budgeted for the school's 2 largest student-led events that garner over 400 attendees combined annually.

Brakes Project Lead – Rose Grand Prix Engineering (GPE) Sept. 2025 – Nov. 2025

- Designed a quick-release solution for RGP012's pedal box.

Skills

Software: MATLAB, Simulink, Python (PyTorch, NumPy/SciPy), C/C++, Java, Arduino, GitHub, Excel
CAD/Sim: SolidWorks, ANSYS Mechanical, ANSYS Fluent, Rocky DEM, NVIDIA Isaac Sim, FEA, CFD
Languages: English (Fluent), Urdu (Conversational), German (Conversational)